

All real phase-retrieving dual frames are generic

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Status. Candidate partial result, likely valid. The real finite-dimensional theorem is complete; the complex case remains open.

Abstract

Given a finite real phase-retrieving frame with arbitrary redundancy, we prove that its phase-retrieving duals form a relatively Zariski-open, Euclidean-open dense, full-measure subset of the affine space of all duals. We give a single explicit polynomial that classifies these duals. Moreover, on the line from any dual to the canonical dual, all but finitely many points do phase retrieval. This removes the source theorem’s minimal-length restriction $m = 2n - 1$.

1 Source question and source limitation

Arabyani-Neyshaburi, Arefijamaal, and Kamyabi-Gol ask whether the phase-retrieving duals of a phase-retrieving frame can be classified and whether they are dense among all duals [1]. The same abstract says that their results establish density only for some frame classes.

Abstract. Recovering a signal up to a unimodular constant from the magnitudes of linear measurements has been popular and well studied in recent years. However, numerous unsolved problems regarding phase retrieval still exist. Given a phase retrieval frame, may the family of phase retrieval dual frames be classified? And is such a family dense in the set of dual frames? Can we present the equivalent conditions for a family of vectors to do weak phase retrieval in complex Hilbert space case? What is the connection between phase, weak phase and norm retrieval? In this context, we aim to deal with these open problems concerning phase retrieval dual frames, weak phase retrieval frames, and specially investigate equivalent conditions for identifying these features. We provide some characterizations of alternate dual frames of a phase retrieval frame which yield phase retrieval in finite dimensional Hilbert spaces. Moreover, for some classes of frames, we show that the family of phase retrieval dual frames is open and dense in the set of dual frames. Then, we study weak phase retrieval problem. Among other things, we obtain some equivalent conditions on a family of vectors to do phase retrieval in terms of weak phase retrieval.

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Their main real density result, Theorem 2.3, treats a frame of exactly $2n - 1$ vectors in $\mathbb{R}^n$.

We also, obtain a sufficient conditions on a family of weak phase retrieval frames to constitute a frame and its canonical dual yields weak phase retrieval, as well.

2. Phase Retrieval Dual Frames

In this section, we address the problem that, given a phase retrieval frame Φ , characterize phase retrieval dual frames of Φ in finite dimensional Hilbert spaces. For some classes of frames we show that phase retrieval dual frames of a given frame are dense in the set of all dual frames. For a frame Φ we denote the set of all its dual frames of Φ by D_Φ and the subset of phase retrieval dual frames is denoted by PD_Φ . We first investigate the relationship between phase retrieval duals of equivalent frames, which is a very useful tool for the main results of this section.

Lemma 2.1. *Suppose that $\Phi = \{\phi_i\}_{i \in I_m}$ is a frame for $\mathcal{H}_n$ and $\mathcal{T}$ is an invertible operator on $\mathcal{H}_n$. Then,*

- (i) $D_{\mathcal{T}\Phi} = (\mathcal{T}^*)^{-1}D_\Phi$.
- (ii) $PD_{\mathcal{T}\Phi} = (\mathcal{T}^*)^{-1}PD_\Phi$.

Proof. It is known that $\mathcal{T}\Phi$ is a frame with $S_{\mathcal{T}\Phi} = \mathcal{T}S_\Phi\mathcal{T}^*$ [13], and so a simple computation assures that $D_{\mathcal{T}\Phi} = \{(\mathcal{T}^*)^{-1}G : G \in D_\Phi\} = (\mathcal{T}^*)^{-1}D_\Phi$. Moreover, (ii) is given as an immediate result of (i) along with Theorem 1.2 $\square$

Remark 2.2. If $\Phi = \{\phi_i\}_{i \in I_m}$ is a frame for $\mathcal{H}_n$ so that $E(\Phi) = k$, which $E(\Phi)$ denotes the excess of Φ , then there exist $i_1, \dots, i_k$ so that $\Phi \setminus \{\phi_{i_j}\}_{j=1}^k$ constitutes a Riesz basis for $\mathcal{H}_n$. Therefore, applying the above notations and without loss of generality, we may consider $\Phi = \{\phi_i\}_{i \in I_n} \cup \{\phi_i\}_{i=n+1}^m$, where $\{\phi_i\}_{i \in I_n}$ is a Riesz basis for $\mathcal{H}_n$. In this point of view, Φ is indeed equivalent to a form as $\{\delta_i\}_{i \in I_n} \cup \{\tilde{\phi}_i\}_{i=n+1}^m$ with redundant elements $\{\tilde{\phi}_i\}_{i=n+1}^m$.

In the next theorem we identify the set of dual frames of a frame $\{\phi_i\}_{i \in I_{2n-1}}$ in n -dimensional real space and show that PD_ϕ is an open and dense subset in D_ϕ .

Theorem 2.3. *Let $\phi = \{\phi_i\}_{i \in I_{2n-1}}$ be a phase retrieval frame in $\mathbb{R}^n$. Then PD_ϕ is an open and dense subset in D_ϕ .*

Proof. First let $\{\phi_i\}_{i \in I_n}$ be the standard orthonormal basis of $\mathbb{R}^n$ and $\phi_j = \sum_{i \in I_n} a_i^j \phi_i$, $j = n+1, \dots, 2n-1$, for some non-zero coefficients $\{a_i^j\}_{i \in I_n}$. Due to the fact that

$$D_\Phi = \left\{ \{S_\Phi^{-1}\phi_i + u_i\}_{i \in I_{2n-1}} : \sum_{i \in I_{2n-1}} \langle f, \phi_i \rangle u_i = 0, \text{ for all } f \in \mathbb{R}^n \right\}$$

by putting $f = \phi_j$, $j = 1, \dots, 2n-1$ we observe that $(2n-1) \times n$ matrix $[u_1 | u_2 | \dots | u_{2n-1}]^T$ is in the null space of G_Φ^T , the transposed Gram matrix of Φ . The fact that,

The result below covers every finite real phase-retrieving frame, with no restriction on its redundancy, and supplies an explicit classification polynomial. It does not address complex frames.

2 Definitions

Let $\Phi = (\phi_i)_{i=1}^m \subset \mathbb{R}^n$ be a frame and let $F = [\phi_1 \cdots \phi_m] \in \mathbb{R}^{n \times m}$ be its synthesis matrix. A frame $G = (g_i)_{i=1}^m$, with synthesis matrix again denoted G , is a dual of Φ when

$$GF^T = I_n. \quad (1)$$

Thus the dual space

$$\mathcal{D}_\Phi = \{G \in \mathbb{R}^{n \times m} : GF^T = I_n\}$$

is a nonempty affine subspace. Its canonical point is

$$G_0 = S_\Phi^{-1}F, \quad S_\Phi = FF^T. \quad (2)$$

A real frame G does phase retrieval when

$$|\langle x, g_i \rangle| = |\langle y, g_i \rangle| \quad (1 \leq i \leq m) \implies x = \pm y.$$

Write $\mathcal{PD}_\Phi$ for the phase-retrieving members of $\mathcal{D}_\Phi$.

For $I \subseteq [m]$, define

$$R_I(G) = \sum_{\substack{J \subseteq I \\ |J|=n}} \det(G_J)^2, \quad (3)$$

where G_J is the $n \times n$ submatrix with columns in J ; an empty sum is zero. Finally set

$$P(G) = \prod_{I \subseteq [m]} (R_I(G) + R_{I^c}(G)). \quad (4)$$

3 Proof intuition

The real complement-property theorem says that phase retrieval is exactly the condition that, for each partition of the frame vectors, at least one side spans. A side spans precisely when at least one maximal minor is nonzero; summing squares of all such minors turns this into one polynomial inequality. Multiplying over all partitions gives (4).

The only way this algebraic certificate could fail to imply density in the affine dual space would be if it vanished identically there. The canonical dual rules that out: it remains phase retrieving because its vectors are an invertible self-adjoint image of the original phase-retrieving frame. The zero set of a nonzero polynomial is nowhere dense and null. On a line toward the canonical dual, the same polynomial becomes a nonzero univariate polynomial, giving only finitely many exceptional points.

4 Main theorem

Theorem 1 (Arbitrary real redundancy). *Let $\Phi = (\phi_i)_{i=1}^m$ be any finite phase-retrieving frame for $\mathbb{R}^n$. Then*

$$\mathcal{PD}_\Phi = \{G \in \mathcal{D}_\Phi : P(G) \neq 0\}, \quad (5)$$

with P given by (4). Consequently $\mathcal{PD}_\Phi$ is relatively Zariski open, Euclidean open and dense in $\mathcal{D}_\Phi$, and its complement has Lebesgue measure zero in that affine space.

More strongly, for every $G \in \mathcal{D}_\Phi$, let

$$G_t = (1 - t)G + tG_0,$$

where G_0 is the canonical dual. Then G_t is a phase-retrieving dual for all real t except a finite set. In particular, every dual is a limit of phase-retrieving duals lying on this line.

Proof. We use the real complement-property characterization of phase retrieval [2, Theorem 2.8]; it is also restated as Theorem 1.4 of the source paper [1, p. 3]. It says that a real frame $G = (g_i)_{i=1}^m$ does phase retrieval if and only if, for every $I \subseteq [m]$, either

$$\text{span}\{g_i : i \in I\} = \mathbb{R}^n \quad \text{or} \quad \text{span}\{g_i : i \in I^c\} = \mathbb{R}^n.$$

By maximal-minor rank detection,

$$R_I(G) > 0 \iff \text{span}\{g_i : i \in I\} = \mathbb{R}^n.$$

Every factor in (4) is nonnegative. Hence $P(G) \neq 0$ exactly when every partition has a spanning side. This proves (5).

Next, G_0 in (2) is a dual by direct multiplication: $G_0 F^T = S_\Phi^{-1} F F^T = I_n$. It also does phase retrieval. Indeed, if

$$|\langle x, S_\Phi^{-1} \phi_i \rangle| = |\langle y, S_\Phi^{-1} \phi_i \rangle| \quad \text{for every } i,$$

self-adjointness of S_Φ^{-1} rewrites this as equality of the Φ -measurements of $S_\Phi^{-1}x$ and $S_\Phi^{-1}y$. Since Φ does phase retrieval, $S_\Phi^{-1}x = \pm S_\Phi^{-1}y$, and invertibility gives $x = \pm y$. Therefore $P(G_0) > 0$.

The restriction of P to the finite-dimensional affine space $\mathcal{D}_\Phi$ is thus a nonzero polynomial. Its nonvanishing locus is relatively Zariski open. The zero set of a nonzero real polynomial has empty Euclidean interior and Lebesgue measure zero, proving the asserted openness, density, and full-measure statement.

Finally, (1) shows that $G_t \in \mathcal{D}_\Phi$ for every t . The function

$$q_G(t) = P((1 - t)G + tG_0)$$

is a real polynomial, and $q_G(1) = P(G_0) > 0$. Thus q_G is not identically zero and has only finitely many real roots. By (5), precisely those roots can fail phase retrieval. Choosing nonroots tending to zero proves the last assertion. $\square$

5 Scope, verification, and novelty

The proof is exact and has no computational dependency. It upgrades source Theorem 2.3 from $m = 2n - 1$ to arbitrary finite m and adds an explicit global classification, a full-measure conclusion, and radial finite-exception genericity. The complex case remains open because the real complement property used above has no direct complex counterpart of this form.

A bounded search on 11 August 2026 checked the run indexes, the exact title, and official arXiv API queries for “phase retrieval dual frames,” “alternate dual frames” with “phase retrieval,” and “dual frames” with “open and dense” and “phase retrieval.” Each exact query returned only arXiv:2301.05045v1. No prior arbitrary-redundancy or radial theorem was found. Novelty is plausible, not an exhaustive certification.

Human-review recommendation. Check the equivalence between (4) and the complement property and the canonical-dual argument. The remaining algebraic-topological consequences are standard.

References

- [1] F. Arabyani-Neyshaburi, A. A. Arefijamaal, and R. A. Kamyabi-Gol, *Characterization of (weak) phase retrieval dual frames*, arXiv:2301.05045v1 (2023).
- [2] R. Balan, P. G. Casazza, and D. Edidin, *On signal reconstruction without phase*, Appl. Comput. Harmon. Anal. 20 (2006), 345–356.